

# why is it still not done?

digitalisierung und programmierung · prof. dr. nicolas meseth

- 1 where the time goes
- 2 three levers
- 3 what it costs to be sure

part 1

# where the time goes

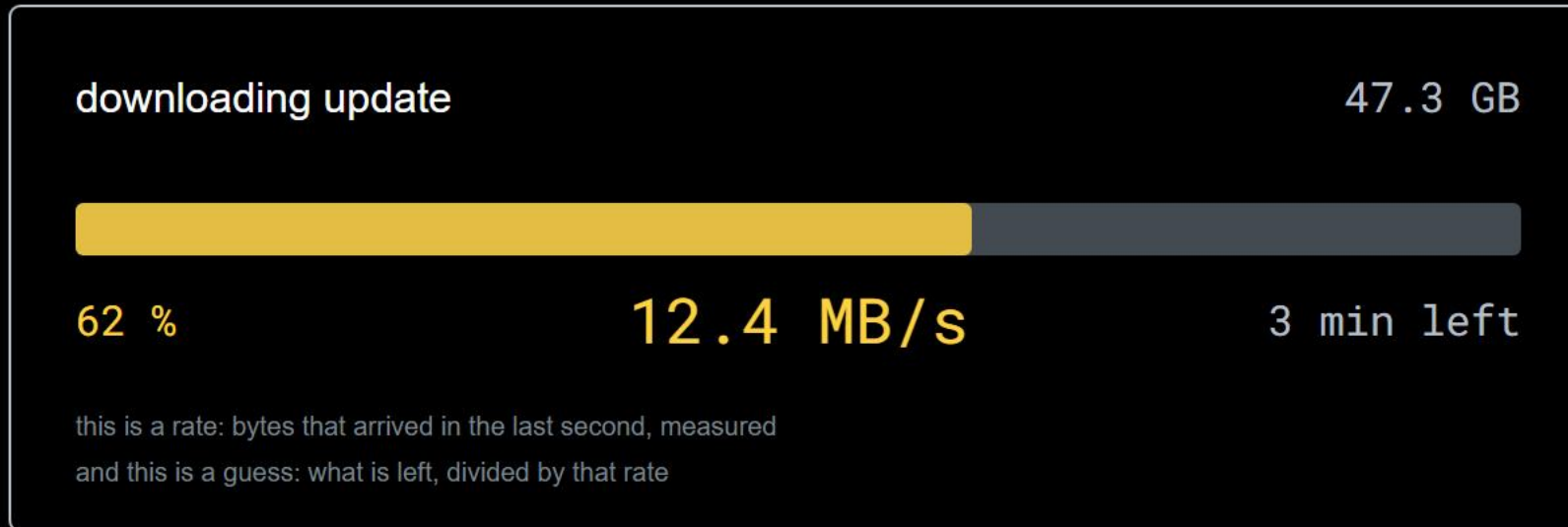
two kilobytes.  
twenty-seven minutes.



image: ai-generated (gpt-image-2)

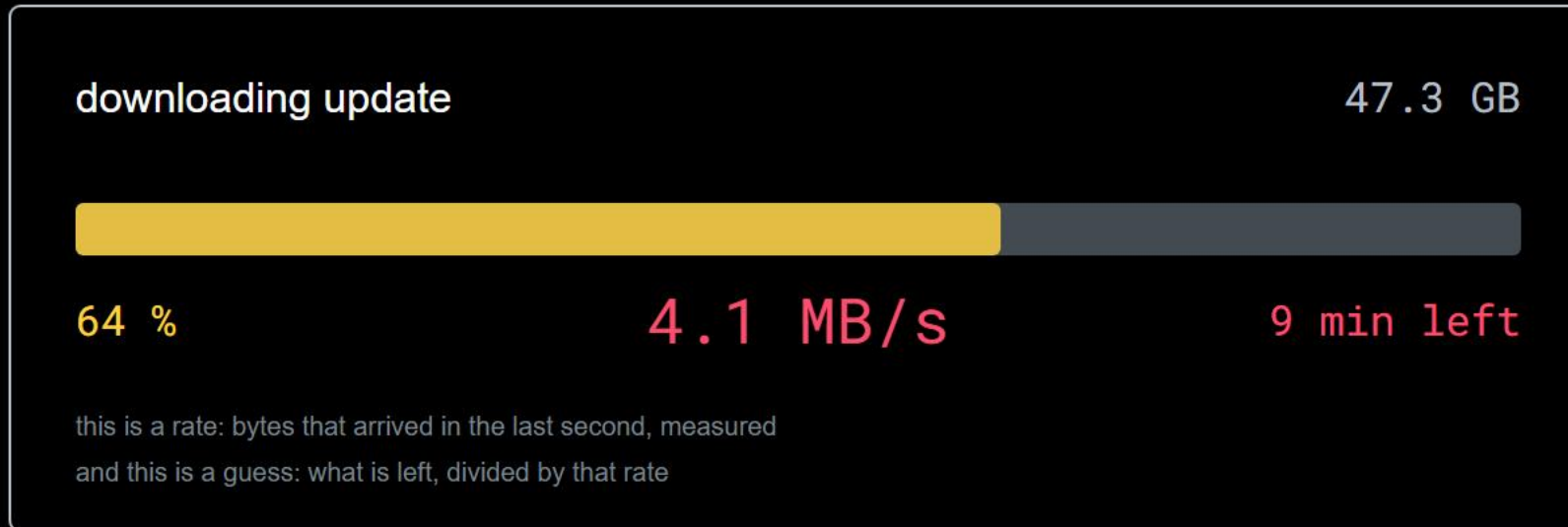
where the time goes

you know this screen



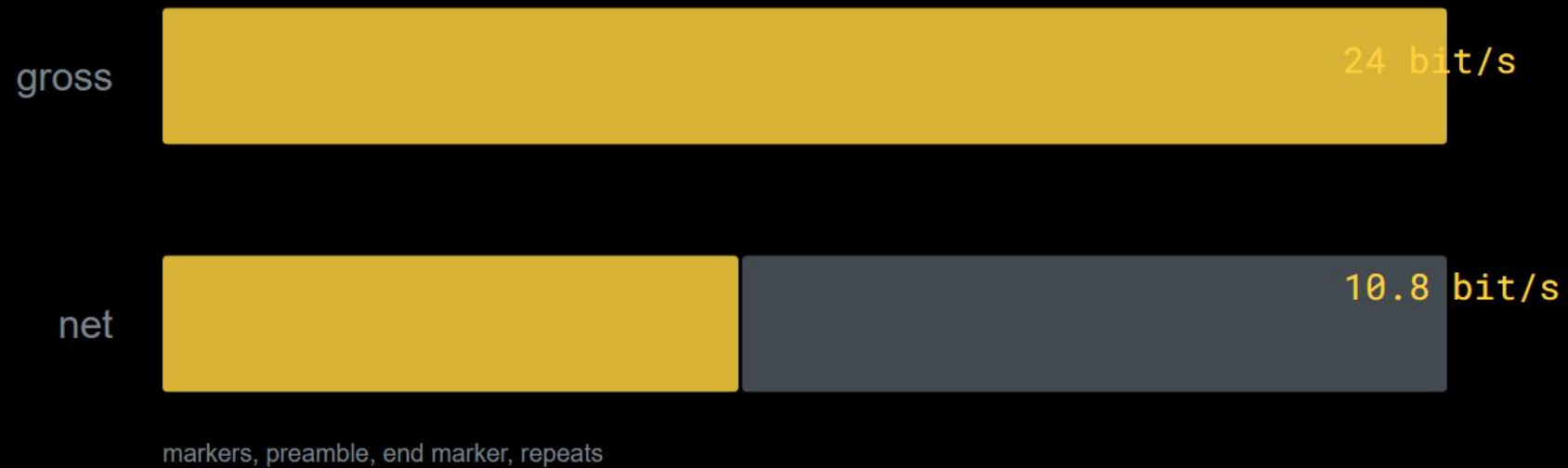
the rate is measured. the minutes are a guess made from it, and they jump when it does.

you know this screen



the rate is measured. the minutes are a guess made from it, and they jump when it does.

gross and net



nothing was lost. most of it was administration.

where the 24 went

every symbol carries data



24.0

deductions multiply. they do not add up.

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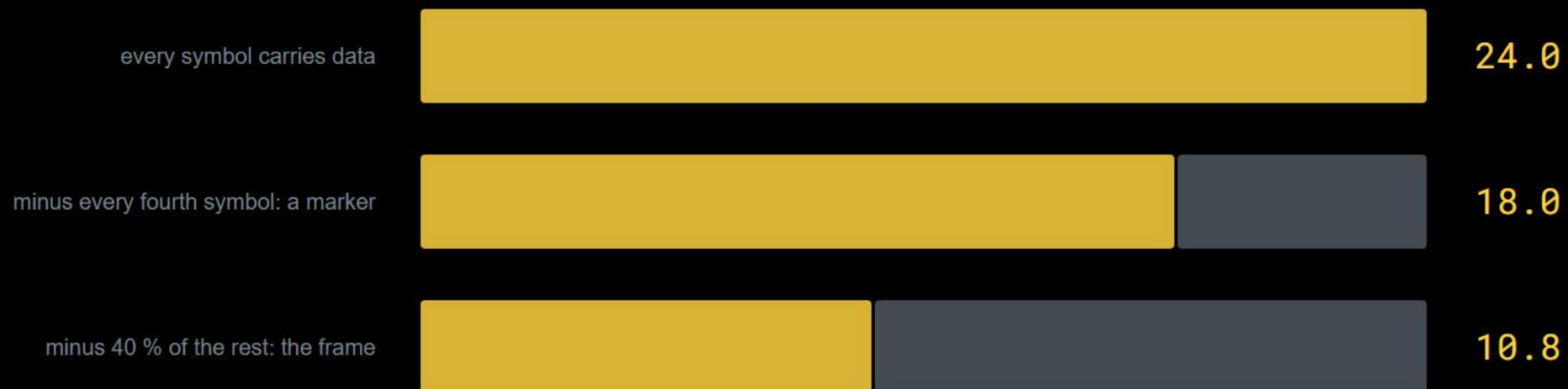
where the time goes

where the 24 went



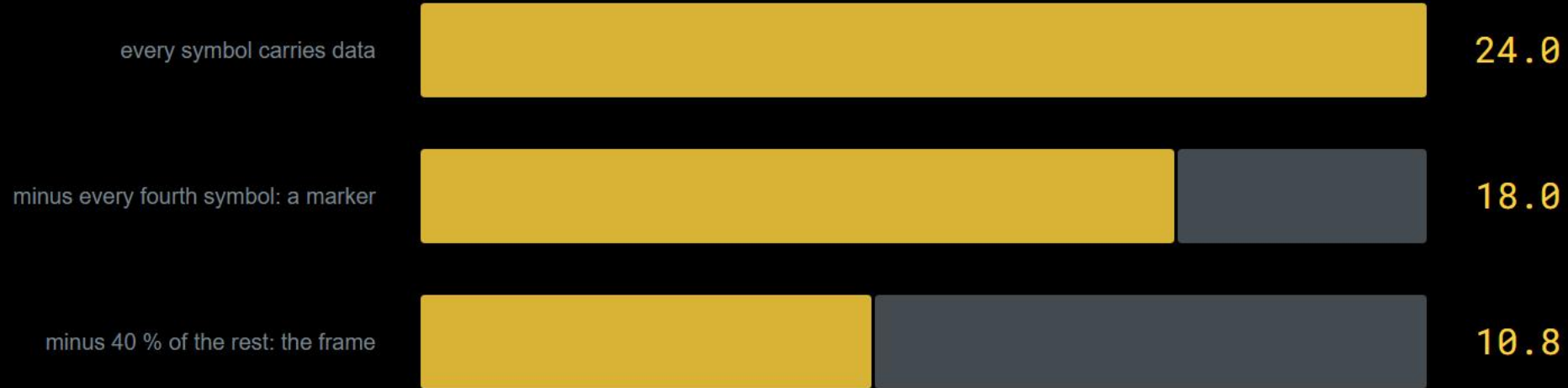
deductions multiply. they do not add up.

where the 24 went



deductions multiply. they do not add up.

where the 24 went



25 % + 40 % = 65 %, so 8.4?  
no. the second cut only applies to what the first left over.

deductions multiply. they do not add up.

pulling the wrong lever

10 symbols per second: a symbol is 100 ms



50 ms window

20 bit/s, and all of it readable

20 symbols per second: a symbol is 50 ms



50 ms window

40 bit/s on paper, nothing readable

a lever past the bottleneck does not help. it destroys.

where the time goes

# your limit is a measurement

not the sensor's datasheet: it does not know your room

not the best team's number: their setup is not yours

not the arithmetic with the values you wish for

what your link really does

| team   | net bit/s | how it was measured                                |
|--------|-----------|----------------------------------------------------|
| team 3 | 21        | 50 messages, raised the rate until the first error |
| team 9 | 14        | same, but by the window in the afternoon           |
| team 6 | 12        | estimated, not measured                            |
| team 1 | 31        | with a cardboard tube around the sensor            |

same hardware. different rooms. different numbers.

part 2

# three levers

three levers, and only three

$$t = n / (r * b)$$

a faster computer is not one of them.

three levers, and only three

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**b** more bits per symbol  
costs safety margin: narrower bands

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three levers, and only three

$$t = n / (r * b)$$

- b** more bits per symbol  
costs safety margin: narrower bands
- r** more symbols per second  
costs the measurement window

a faster computer is not one of them.

three levers, and only three

$$t = n / (r * b)$$

- b** more bits per symbol  
costs safety margin: narrower bands
- r** more symbols per second  
costs the measurement window
- n** fewer bits in total  
costs a few milliseconds of computing

a faster computer is not one of them.

the same 20 bit/s, twice

team a

$$10 * 2$$

10 symbols per second, 4 colours

**20 bit/s**

close to the measurement window

team b

$$5 * 4$$

5 symbols per second, 16 colours

**20 bit/s**

close to the scatter limit

the product decides. the two factors decide what breaks first.

# the third lever has its own deck

fewer bits in total: compression

costs milliseconds, not safety margin

works before the link, so it works on any link

next: [compression](#)

part 3

# what it costs to be sure

three levers, one file

as it is



27 min

16 000 bits, 5 symbols per second

the levers are independent. that is why they multiply.

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what it costs to be sure

three levers, one file



16 000 bits, 5 symbols per second

the levers are independent. that is why they multiply.

three levers, one file



16 000 bits, 5 symbols per second

the levers are independent. that is why they multiply.

three levers, one file



16 000 bits, 5 symbols per second

the levers are independent. that is why they multiply.

every tenth one again



nothing was lost. the link was busy anyway.

acknowledge each, or a block

acknowledge every message



acknowledge every block of ten



yellow: payload · dark: waiting for the acknowledgement

but one damaged message costs the whole block again

how big a block? that depends on a number you have to measure.

# throughput is not delay

20 bit/s does not mean a one-byte command is there after 0.4 seconds. around it sit the frame, the recognising and the reaction.

today

- 1 measure your own net rate, and write down how
- 2 find the lever that is cheapest on your setup, with the number that says so
- 3 pick a block size for your acknowledgements, with a reason

the neck decides  
how much gets through.  
you decide  
what you send.



image: ai-generated (gpt-image-2)

what it costs to be sure